Emergency Vehicle Sensing and Communication Network (ISAC-V2X)

An ISAC-V2X Framework for Traffic-Density Sensing and Emergency Vehicle Alerting

Manas Pandya
Anant Narayan
Maitrey Dave
Aniruddha Wairagade
Datt Suthar

ECE310 – Wireless Communications
Ahmedabad University
Milestone 2 Evaluation Report

CONTENTS

I	Introduction	2
II	Problem Definition	2
III	Proposed System	2
IV	State of the Art and Related Work	3
V	Research Gap and Project Novelty	4
VI	Scope, Assumptions, and Constraints	4
VII	Criteria for Assessing the Proposed System	4
VIII	Research Methodology	5
IX	Milestone Plan and Expected Deliverables	5
X	Conclusions and Future Outlook	5
	References	6

LIST OF FIGURES

1	Proposed ISAC-V2X system architecture	3
2	Closed-loop operating sequence	3

LIST OF TABLES

I	Summary of related work and project positioning	4
II	Project evaluation criteria	5
III	Milestone roadmap	5

Milestone 2: Problem Definition, Literature Review, SOTA Positioning, and Scope Finalization.

***Abstract*—**Emergency vehicles often experience delays as a result of the traffic congestion on the routes that they are required to travel to reach their destinations. Additionally, the drivers of the vehicles that are in the vicinity of the emergency vehicle may not be aware of the presence of such an emergency vehicle until the vehicle is at a very close range. To address these issues, the present project proposes the development of an ISAC-based V2X communication framework that utilizes a shared radio signal to both measure the traffic density of the route to be traveled by the emergency vehicle and to provide V2X communication from the emergency vehicle to the other vehicles on the road. Roadside units are proposed to be installed that will allow for the measurement of the number of emergency vehicle beacon signals that are present on the road, which will provide an indication of the traffic density. This information will then be used by a route optimization module based on Dijkstra or A^* to select a less-congested path.

For the second milestone of the project, a simulation-based approach will be utilized that utilizes an OpenISAC-inspired continuous-wave OFDM physical layer for the communication between the emergency vehicle and the roadside units. While the overall system architecture maintains that the emergency vehicle initiates the communication and the roadside units respond with the route optimization feedback prior to the emergency vehicle broadcasting the alert, the second milestone will instead focus upon simulating the communication between the emergency vehicle and the roadside units in code. This simulation will demonstrate the concept of the proposed system and make the code visible to the project team, temporarily bypassing the physical measurement of the traffic density on the road. Additionally, the simulation will establish the performance of the proposed physical layer. Hardware validation and the implementation of the physical measurement of traffic density will be deferred to the third milestone of the project.

Index Terms: Integrated sensing and communications, ISAC, V2X, emergency vehicle, OpenISAC, 2.4 GHz, traffic density, route optimization.

***Index Terms*—**Integrated sensing and communications, ISAC, V2X, emergency vehicle, DSRC, software-defined radio, GNU Radio, USRP B210, OFDM, traffic density, route optimization.

I. Introduction

Emergency response systems are dependent upon an emergency vehicle (EV) being able to move through the traffic without unnecessary delay. In many cases, the driver of the EV may not have sufficient information regarding the density of each of the road segments in the route that the EV needs to take to reach the emergency as quickly as possible. Furthermore, the vehicles in the traffic may not receive an early warning that an EV is approaching their road segment.

There are a variety of emergency-vehicle warning systems that have been developed and are in use today. For example, HAAS Alert provides responders-to-vehicles (R2V) and responders-to-responders (R2R) alerts that are based upon real-time vehicle information and cloud-based processing of that information. As such, the system is dependent upon the vehicles being connected, the use of navigation applications, GPS, cellular communications, and cloud processing of the information.

In order to develop a system that integrates sensing and V2X communication natively into an emergency vehicle warning system, this project utilizes the OpenISAC platform, which is an open-source OFDM-ISAC platform that was originally developed for USRPs. By modifying the frame structure of OpenISAC, a custom V2X handshake was simulated that integrated emergency vehicle alerting and density-based route optimization on a shared physical link.

II. Problem Definition

The project problem is defined by three related operational challenges:

A. Emergency-vehicle response latency

An EV may lose critical response time when it enters a congested road corridor. Without a live estimate of the density of alternative road segments, the driver or routing system cannot reliably determine whether the current route remains preferable.

B. Route congestion blindness

The EV requires a mechanism that converts radio observations into a useful estimate of traffic density. The proposed project does not attempt to reconstruct the complete physical traffic state. Instead, it uses the number of active beacon signals observed on a road segment as a low-cost density proxy. This proxy is then used to update the cost of the corresponding road segment in a routing graph.

C. Belated civilian awareness

Civilian drivers travelling along the EV's selected route may become aware of the emergency vehicle only when visual or audible warning cues become noticeable. The proposed system therefore includes a direct RF broadcast mechanism through which the EV can send a targeted "make way" alert to nearby nodes.

D. Core problem statement

The project seeks to design and validate a lightweight two-way RF system in which a shared physical link supports both traffic-density sensing and V2X communication. The intended closed loop is:

$$\begin{aligned} \text{EV intent} &\rightarrow \text{RF density sensing} \\ &\rightarrow \text{route selection} \\ &\rightarrow \text{RF alert broadcast.} \end{aligned} \tag{1}$$

The shared-channel system is the centerpiece of our project.

III. Proposed System

A. System concept

The proposed architecture has three parts: a density-sensing module, a route-optimization module and an alert handshake module. The roadside-unit (RSU) and relay nodes use beacon signals, for each road segment to get a rough idea of traffic density. To show how this works in code for Milestone 2 we simulate the communication loop. The OBU sends a route request to the RSU. The RSU then acknowledges that request and sends back route feedback. This feedback is based on Dijkstra or A^* calculations.

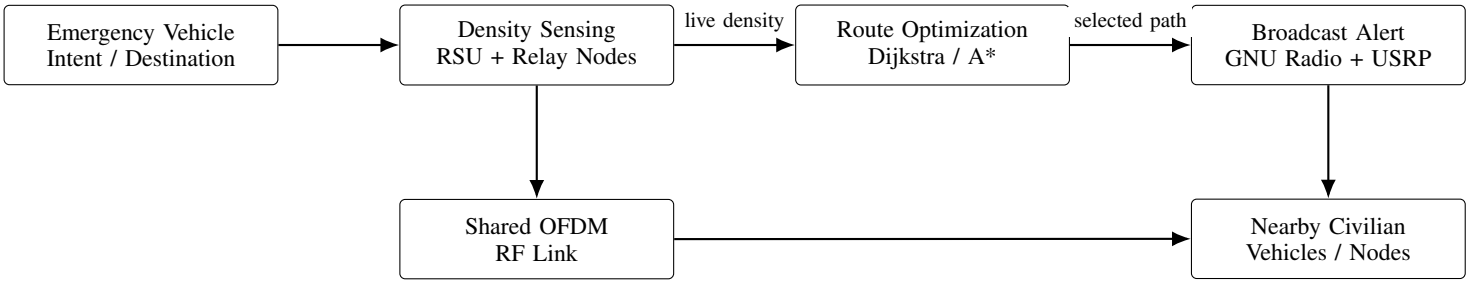


Fig. 1. Proposed ISAC-V2X system architecture showing the density sensing, route optimization, and broadcast alert modules.

B. Density sensing module

The density module is based on beacon counting. Each road segment is associated with one or more sensing/relay nodes. The number of active beacon signals observed during a measurement interval is treated as a proxy for the number of vehicles present on that segment. This approach is deliberately lightweight: the project does not require a dedicated radar sensor or detailed object classification at the Milestone 2 stage.

The density proxy will later be validated against manually counted ground-truth traffic in at least two laboratory test scenarios; this is one of our main objectives.

C. Route optimization module

The routing module converts the measured density of each road segment into a dynamic edge cost. The project plan identifies Dijkstra and A* as the candidate routing algorithms. The selected algorithm will recompute the emergency vehicle path when live density information changes.

The routing layer is intentionally separated from the RF physical layer so that it can first be tested using synthetic or recorded density values. This modular structure reduces the difficulty of debugging the complete system and permits independent verification of the routing logic.

D. Alert and Handshake module

The alerting mechanism works as a REQUEST ACK and FEEDBACK handshake between the EV and the RSU. The OBU starts the conversation by asking for route clearance. The RSU takes the request sends an ACK and gives the FEEDBACK that holds the optimized route instructions. If a timeout happens the system retries to keep things reliable. The target packet delivery ratio (PDR) stays at 90%. We check this only in simulation, for Milestone 2.

E. Physical-layer concept

The physical layer uses an OpenISAC-inspired continuous-wave OFDM architecture (transmitting frames back-to-back, similar to LTE or NR) rather than burst-based Wi-Fi-style packets. This uniform timing helps stabilize sensing functionality. The frame structure consists of one full-band Zadoff-Chu synchronization symbol per frame, pilot subcarriers in every data symbol, and QPSK data mapped with LDPC coding.

The receiver operates in two states: `SYNC_SEARCH` and `NORMAL`, with synchronization detected via a normalized correlation and a peak-to-average threshold. While inspired by NR, this PHY is not strictly standard-compliant. The simulation is configured for 2.4 GHz, matching OpenISAC's verified synchronization tests at this frequency. Deployments at 5.8 or 5.9 GHz are reserved as future hardware options.

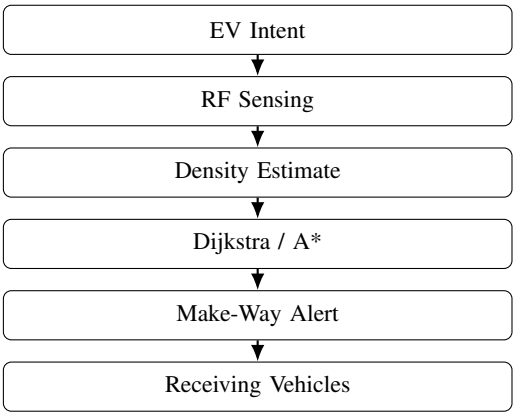


Fig. 2. Closed-loop operating sequence of the proposed ISAC-V2X system.

TABLE I
MILESTONE 2 PHY SIMULATION PARAMETERS

Parameter	Value
Carrier	2.4 GHz
Bandwidth / sample rate	10 MHz
Subcarriers N	1024 (spacing $\sim$9.77 kHz)
Cyclic prefix	128 samples (12.8 μs)
Symbols per frame M	100 (frame $\sim$11.5 ms)
Modulation / coding	QPSK, LDPC
Channel model	Noise, delay, frequency and timing offset; Doppler set to 0

IV. State of the Art and Related Work

Milestone 2 uses three representative benchmarks to position the proposed system: HAAS Alert, DSRC-based rear-end collision warning, and ISAC-enabled vehicular networks. The comparison is summarized in Table II.

A. HAAS Alert

The HAAS Alert report describes two main use cases: responder-to-vehicle (R2V) alerts for civilian drivers and responder-to-responder (R2R) alerts between equipped emergency vehicles [1]. In the R2V case, alerts are associated

TABLE II
SUMMARY OF RELATED WORK AND PROJECT POSITIONING.

Benchmark	Main capability	Strength	Limitation relative to this project
HAAS Alert [1]	R2V and R2R emergency alerts using real-time vehicle data, cellular connectivity, and cloud processing.	Demonstrates operational EV warning concepts and evaluates usability.	Does not combine RF traffic-density sensing with the alerting loop; depends heavily on cloud/cellular connectivity.
OpenISAC *et al.* [2]	Real-time continuous-wave OFDM-ISAC platform evaluated on USRPs.	Provides a robust, open-source foundation for shared waveform processing.	Downlink-focused; lacks a dedicated vehicular request/acknowledge handshake layer.
Proposed System	OpenISAC-inspired PHY model + dynamic routing + REQUEST/ACK/FEEDBACK handshake over shared OFDM link.	Combines sensing proxies, routing, and robust end-to-end V2X handshake into one closed loop.	Simulation-only for Milestone 2; hardware sensing deferred to M3.

with the predicted path of an emergency vehicle and can be delivered through navigation applications or vehicle systems. In the R2R case, the system identifies potential collisions between emergency vehicles connected to the HAAS Safety Cloud.

The system therefore establishes an important benchmark for the alerting component of the proposed project. Its evaluation also demonstrates that operational suitability includes more than communication accuracy: alert timing, usability, configuration, reporting, and operator interaction are relevant considerations.

B. OpenISAC

OpenISAC provides an open-source, real-time OFDM-ISAC processing framework. While existing implementations demonstrate successful synchronization and continuous-wave OFDM processing on USRPs, they are primarily focused on standard physical layer ISAC performance metrics rather than targeted V2X application layers.

V. Research Gap and Project Novelty

While open OFDM-ISAC platforms exist, the reference architecture we build upon is fundamentally downlink-focused and does not feature a dedicated vehicular request/acknowledge handshake or route-feedback application layer. Our contribution is the development of this bidirectional handshake layer, run end-to-end over a shared continuous-wave frame. By isolating this loop in simulation, we bridge the gap between low-level RF sensing and a closed-loop dynamic routing application.

Code Attribution: The simulation environment uses an OpenISAC-inspired PHY model. Our custom routing and handshake logic (BSD-2-Clause) is distinct from the core OpenISAC reference architecture (CC BY 4.0).

VI. Scope, Assumptions, and Constraints

A. Scope

The scope finalized for the project is:

- Implement an OpenISAC-based PHY model in simulation
- Isolate and simulate the REQUEST/ACK/FEEDBACK handshake between the OBU and RSU
- Execute dynamic route selection (Dijkstra/A*) using synthetic denisty inputs.
- Measure simulated PDR, ACK success, and retry statistics.
- Implement Dijkstra/A* based dynamic route selection;
- Transmit emergency “make way” alerts to vehicles/nodes on the selected route;
- Support at least three receiving nodes in the RF demonstration;
- Identify EV alert signals using a preamble/header and evaluate false-positive and false-negative behaviour;
- Integrate the modules into an end-to-end laboratory demonstration.

B. Assumptions

For Milestone 2, the following assumptions are made:

- Vehicles are quasi-static within a single frame duration.
- Doppler shift is neglected (assumed to be 0Hz) as it is small relative to the subcarrier spacing.
- The wireless environment is represented via a synthesized channel model (noise, delay, frequency/timing offsets).
- Traffic density inputs are synthetically generated for the M2 routing logic.
- RF communication can be tested using a controlled laboratory topology;
- Virtual relay nodes can be used when physical nodes are insufficient for early testing;
- The project’s functionality is evaluated as a prototype rather than a production-certified emergency response system.

C. Constraints

We haven’t validated any real hardware(USRP) in this Milestone 2. Also channel conditions are simplified (0 Hz Doppler). Currently our simulation is also bound by OpenISAC’s host-processing limits.

VII. Criteria for Assessing the Proposed System

The project requires measurable criteria so that later milestones can determine whether the proposed architecture is viable. Table III defines the principal criteria.

TABLE III
PROJECT EVALUATION CRITERIA.

Criterion	Target / assessment method
RF reliability	Demonstrate at least 90% PDR across receiving nodes.
Density proxy validity	Compare beacon-count estimates with manual ground truth in laboratory scenarios.
Route adaptation	Verify that the Dijkstra/A* recomputes path when synthetic density changes.
End-to-end operation	Demonstrate EV intent → synthetic sensing → routing feedback → alert broadcast.
Signal identification	Measure false-positive and false-negative behaviour of the sync threshold..
Latency	Compare alert latency and time saved.

TABLE IV
MILESTONE ROADMAP.

Phase	Focus	Expected output
Milestone 1	Problem and SOTA positioning	Problem definition, literature review, SOTA comparison, research gap, scope, assumptions, evaluation criteria, and finalized architecture.
Milestone 2	Code-only Simulation	OpenISAC-inspired PHY simulation, handshake logic, synthetic routing tests, BER/BLER curves, threshold sweeps.
Milestone 3	System Integration	Hardware validation (USRP B210), real-world density validation, latency and time-saved metrics.
Final phase	Evaluation and report	Final documented PHY parameters, signal-ID results, logs, and defense.

Our core objectives include PDR, density-proxy validation, dynamic routing, end-to-end integration, and signal identification. Performance comparisons of sensing-enabled and non-sensing cases are also included among the expected deliverables.

VIII. RESEARCH METHODOLOGY

Milestone 2 is primarily a design and research milestone. Our methodology therefore focused on establishing a traceable link between the identified problem, the literature, the proposed architecture, and the measurable criteria.

A. Literature review

The literature review uses four benchmark categories. The HAAS Alert report was used to understand an operational emergency-vehicle warning architecture and its evaluation dimensions. The DSRC study was used to understand direct RF vehicle-safety communication and the role of transmission delay and positioning errors. The ISAC review was used to establish the architectural principle of jointly using radio resources for sensing and communication. The C-V2X and DSRC performance analysis study was used to make decisions about the standard used for transmission and the performance limitations our system will have to work around.

B. System decomposition

The proposed system is decomposed into independent modules:

1) density sensing,
2) route optimization,
3) RF broadcast alerting,
4) signal identification, and
5) end-to-end integration.

This decomposition permits each software or DSP block to be tested independently before the final integration stage.

C. Density validation methodology

Because physical density validation via manual vehicle counting cannot be executed in code, the Milestone 2 simulation relies on synthetic density data to validate the algorithm's decision-making and route-switching logic. Real-world validation against physical manual ground truth is deferred to Milestone 3.

D. Simulation methodology

The Milestone 2 evaluation is entirely software-driven. PDR is measured by simulating multiple packet transmissions across varying SNR points to generate BER/BLER curves, which are evaluated against standard OpenISAC performance metrics. The REQUEST/ACK handshake is evaluated by tracking ACK success rates and analyzing retry statistics under different simulated channel impairments. Furthermore, we conduct a synchronization threshold sweep to measure false-positive and false-negative detection rates during the `SYNC_SEARCH` state.

E. Route and alert validation

The route module will first be tested using generated density maps. Once the route logic is stable, it will receive live or recorded density updates. The selected route will then determine the intended alert recipients. The RF layer will be evaluated using packet delivery ratio, signal identification accuracy, and alert latency.

IX. MILESTONE PLAN AND EXPECTED DELIVERABLES

The project roadmap divides development into four phases. Milestone 2 is the current phase and focuses on problem definition, literature review, state-of-the-art positioning, and scope finalization.

The expected technical deliverables include GNU Radio TX/RX flowgraphs, a standalone density and routing module, a 3+ node SDR testbed, performance benchmarks, documented PHY specifications, signal-identification results, and threshold-based alerting logs.

X. CONCLUSIONS AND FUTURE OUTLOOK

Milestone 2 creates a tested software base for an ISAC-V2X emergency vehicle sensing and communication prototype. The project looks at three problems: delays in emergency response no live information about how busy roads are and slow alerts to people who are not involved. The solution is a loop that

works both ways of sending messages one way. It uses a two-way REQUEST/ACK/FEEDBACK V2X process along with changing how routes are chosen. The review of what's already available shows that current systems have useful parts but there is a need for complete solutions that work at the edge with radio frequency. HAAS Alert shows how to warn emergency vehicles from the cloud and let responders send alerts to each other. OpenISAC gives a way to process data in real time using continuous-wave OFDM-ISAC. The project brings these together by building a part for vehicles on top of a physical layer inspired by OpenISAC. This allows route feedback over a shared radio signal. By using a software-simulation at 2.4 GHz for this step the project made sure the main communication and deciding part worked well. Testing with data made it easy to check the Dijkstra and A^* routing methods as well as how well the OBU and RSU parts work together without dealing with real hardware problems. This way of testing gave numbers like how many messages get through how often they are accepted and how well the system knows when things are in sync. The main plan for Milestone 3 is to keep the separate parts as the project moves from software to real equipment. Since the way routes are chosen and the radio part have been tested in code any problems during setup can be found and fixed step by step. At the end of Milestone 2 the project has a tested part, for vehicles, a simulated radio part and a clear way to use USRP radios to check how the whole system works in real life. This includes checking how well it deals with moving signals how accurately it measures roads and how fast it works.

REFERENCES

[1] M. Monetti, R. Casparro, K. Dooley, and T. Mackanin, *Emergency Vehicle Warning System (HAAS Alert): Virtual Technology Demonstration Report*, National Urban Security Technology Laboratory, U.S. Department of Homeland Security, Science and Technology Directorate, July 2021.

[2] X. Zhao, S. Jing, F. Hui, R. Liu, and A. J. Khattak, "DSRC-based rear-end collision warning system – An error-component safety distance model and field test," *Transportation Research Part C: Emerging Technologies*, vol. 107, pp. 92–104, 2019, doi: 10.1016/j.trc.2019.08.002.

[3] H. I. Abbasi, R. Gholmieh, T. V. Nguyen, S. Patil and J. Misener, "LTE-V2X (C-V2X) Performance in Congested Highway Scenarios," *ICC 2022 - IEEE International Conference on Communications*, Seoul, Korea, Republic of, 2022, pp. 303-308, doi: 10.1109/ICC45855.2022.9838706.

[4] Z. Du et al., "Toward ISAC-Empowered Vehicular Networks: Framework, Advances, and Opportunities," in *IEEE Wireless Communications*, vol. 32, no. 2, pp. 222-229, April 2025, doi: 10.1109/MWC.002.2300496.